GSE 518
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Problem Set 8

You must turn in problems: 1, 2(a) and (b), 3(a), 4(a) to 4(d), and 5. However, you are strongly encouraged to work on all the problems before the final exam, since the material they cover will be part of it. Please let me know if you have any questions.

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from each of the following distributions. In each case find the method of moments estimator $\widehat{\theta}_{MOM}$ of θ_0.

   (a) A $Gamma(\alpha_0, \beta_0)$ distribution. You can take as given that the mean and variance of a $Gamma(\alpha_0, \beta_0)$ distribution are $E(X) = \alpha_0\beta_0$ and $Var(X) = \alpha_0\beta_0^2$. (*General knowledge:* There is no explicit formula for the MLE estimators of α_0 and β_0, and therefore, the MLE estimators must be obtained through numerically maximization. As you will see, the MOM estimators do have a closed form solution.)

   (b) A distribution with pdf

   $$\begin{aligned} f(x|\theta_{01}, \theta_{02}) &= \frac{1}{2\theta_{02}\sqrt{3}}, \quad \theta_{01} - \theta_{02}\sqrt{3} < x < \theta_{01} + \theta_{02}\sqrt{3} \\ &= 0 \qquad\qquad \text{otherwise.} \end{aligned}$$

   with $-\infty < \theta_{01} < \infty$ and $0 < \theta_{02} < \infty$. (*Hint:* You can take as given that the mean and variance of this density are $E(X) = \theta_{01}$ and $Var(X) = \theta_{02}^2$.)

2. Let $X_1, X_2, \ldots, X_n$ be a random sample from each of the distributions having the following probability density/mass functions. In each case derive the maximum likelihood estimator $\widehat{\theta}_{MLE}$ of θ_0.

   (a) $f(x|\theta_0) = \frac{\theta_0^x e^{-\theta_0}}{x!}$, for $x = 0, 1, \ldots$ and zero otherwise, for $\theta_0 \geq 0$.

   (b) $f(x|\theta_0) = \frac{1}{\theta_0} e^{-(x/\theta_0)}$, for $0 < x < \infty$, and zero otherwise, for $0 < \theta_0 < \infty$.

   (c) $f(x|\theta_0) = \frac{1}{2} e^{-|x-\theta_0|}$, for $-\infty < x < \infty$, and zero elsewhere, and $-\infty < \theta_0 < \infty$. (*Note:* for this particular density do not bother in deriving the $\widehat{\theta}_{MLE}$ since it involves the use of "order statistics", which we have not covered. However, I do want you to derive the Log-likelihood function. I will add some *"general knowledge"* comments about this problem to the answer key.)

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from each of the distributions having the following probability density/mass functions. In each case find the method of moments estimator $\widehat{\theta}_{MOM}$ and the maximum likelihood estimator $\widehat{\theta}_{MLE}$ of θ_0.

   (a) $f(x|\theta_0) = \theta_0 x^{\theta_0 - 1}$, for $0 < x < 1$, and zero otherwise, for $0 < \theta_0 < \infty$. You can take as given that the mean of this density is $E(X) = \frac{\theta_0}{\theta_0 + 1}$.

(b) $f(x|\theta_0) = \frac{1}{\theta_0}$, for $0 < x < \theta_0$ and zero otherwise. You can take as given that the mean of a $Uniform(0, \theta_0)$ is $E(X) = \theta_0/2$. (*Note:* This is the pdf of a $Uniform(0, \theta_0)$. For this particular density do **not** find the maximum likelihood estimator, since it is not straightforward. If *"just for fun"* you want to try to find $\widehat{\theta}_{MLE}$, you can consider the following hints: Let $1\{A\}$ be an indicator function, which equals one if the event A is true and zero otherwise. Then, we can write the pdf of X_i as: $f(x_i|\theta_0) = \frac{1}{\theta_0} \cdot 1\{x_i \leq \theta_0\}$. You can use this to derive the Log-likelihood function. However, you will see that you cannot use calculus to obtain $\widehat{\theta}_{MLE}$).

4. Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean zero and unknown variance σ_0^2, $N(0, \sigma_0^2)$. You can take as given that the pdf of a $N(0, \sigma_0^2)$ distribution is given by $f(x|\sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\left(x^2/2\sigma_0^2\right)}$.

   (a) Obtain the MLE estimator of σ_0^2.

   (b) Obtain the score function.

   (c) Verify that the Score Identity holds.

   (d) Verify that the Information Matrix Identity holds. (*Hint:* It may be helpful to know that if $X \sim N(0, \sigma_0^2)$, then $E(X^4) = 3\sigma_0^4$.)

   (e) Verify that σ_0^2 maximizes $E[\ln f(X|\sigma^2)]$ with respect to σ^2.

5. Suppose the pairs of observations $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ form a random sample from the bivariate normal joint pdf (see Definition 4.5.10 in CB). Derive the estimators for the parameters μ_x, μ_y, σ_x^2, σ_y^2, and ρ using the method of moments and the first two moments of the data. (*Hint:* You can take as given that $E(XY) = \mu_x\mu_y + \rho\sigma_x\sigma_y$, and you may want to use some nice properties you know about bivariate normal random variables (e.g., see p.175 in CB). You may also want to consider the sample moment: $\frac{1}{n}\sum_{i=1}^{n} X_i Y_i$. By the way, this problem is easier than it looks, so do not complicate yourself too much.) *General knowledge (i.e., do not do what follows):* you could also derive the MLE estimator using the pdf in definition 4.5.10. The algebra is messy, but at the end one finds that $\widehat{\theta}_{MLE} = \widehat{\theta}_{MOM}$.